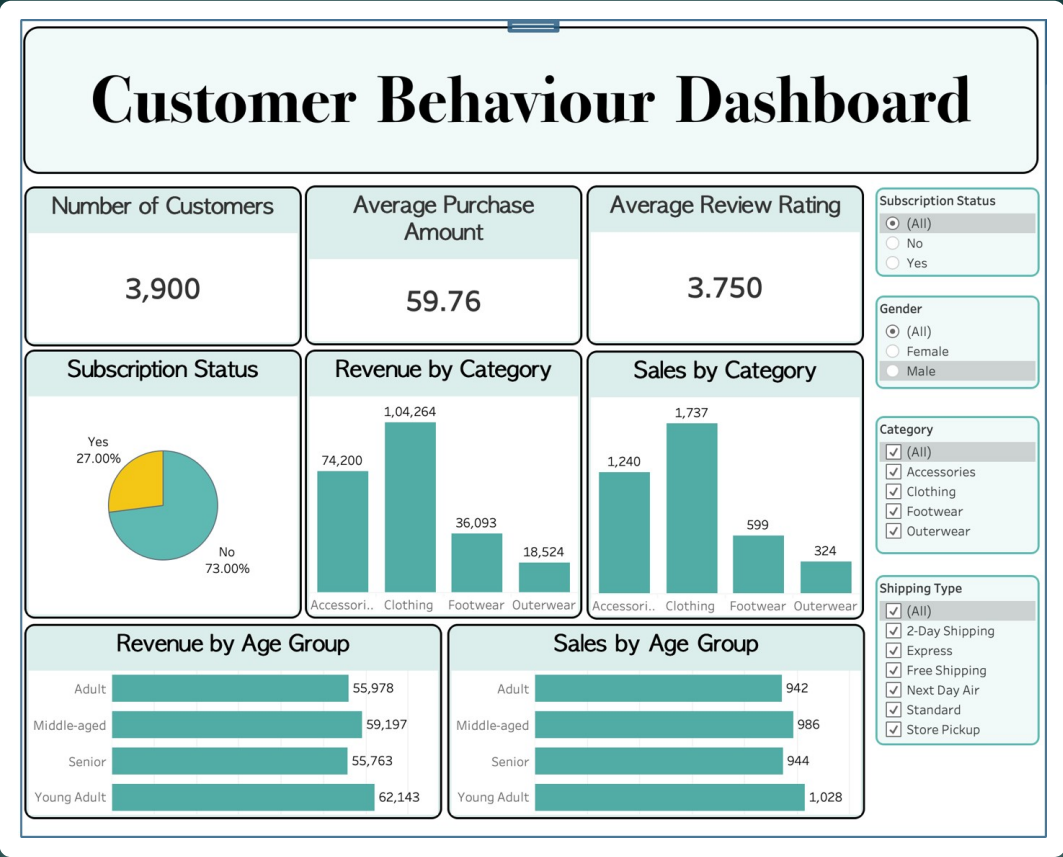


Customer Shopping Behavior Analysis

An end-to-end data analytics project

Python → PostgreSQL → Tableau



Business Problem

THE KEY QUESTION

How can shopping data reveal trends to optimize marketing and product strategy?

Objective

Understand customer shopping behavior to lift sales, engagement, and long-term loyalty.

Focus Areas

- Discounts
- Reviews
- Seasonality
- Payment preferences
- Demographics

Approach

```
#create a column_age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q = 4, labels = labels)
```

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

1

Python (pandas)

Cleaned data, imputed missing review ratings by category median, and engineered age_group and purchase-frequency features.

2

PostgreSQL

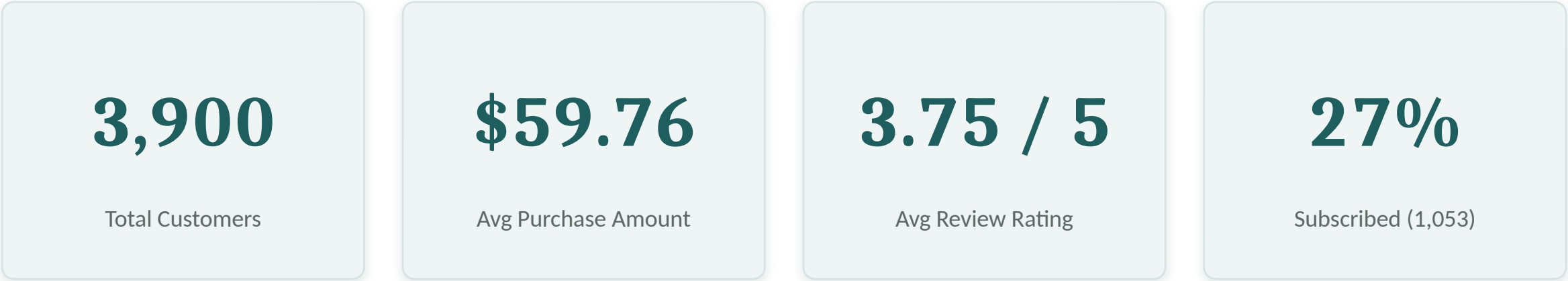
Loaded the data and answered 10 business questions using aggregates, subqueries, CASE, CTEs and window functions.

3

Tableau

Built a live-connected interactive dashboard with KPIs, category & age breakdowns, and cross-filters.

KPIs at a Glance



As shown on the live Tableau dashboard

Number of Customers	Average Purchase Amount	Average Review Rating
3,900	59.76	3.750

73% not subscribed

2,847 of 3,900 customers — the biggest growth opportunity.

Where the Revenue Comes From



Revenue by Category

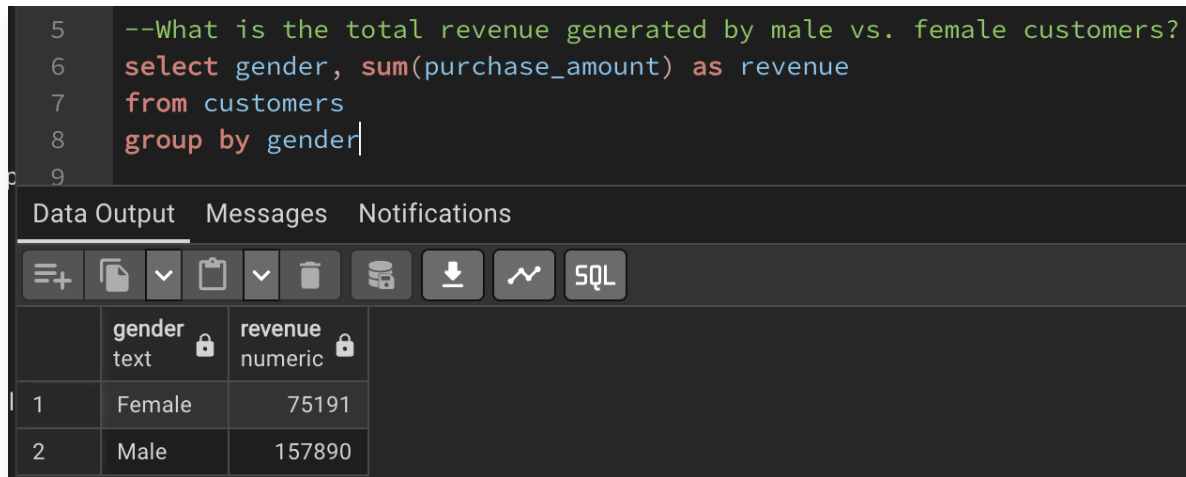
- Clothing — \$104,264 (1,737 orders)
- Accessories — \$74,200 (1,240 orders)
- Footwear — \$36,093 (599 orders)
- Outerwear — \$18,524 (324 orders)

Revenue by Age Group

Young Adult \$62,143 · Middle-aged \$59,197 · Adult \$55,978 · Senior \$55,763

Insight Clothing is the core category (~45% of revenue and orders); revenue is spread evenly across all age groups.

The Demographics Story



The screenshot shows a SQL IDE interface. The top pane contains a SQL query: `--What is the total revenue generated by male vs. female customers?`, `select gender, sum(purchase_amount) as revenue`, `from customers`, and `group by gender`. The bottom pane shows the results in a table with columns 'gender' (text) and 'revenue' (numeric). The results are: 1 Female with revenue 75191, and 2 Male with revenue 157890.

```
5  --What is the total revenue generated by male vs. female customers?
6  select gender, sum(purchase_amount) as revenue
7  from customers
8  group by gender
9
```

	gender text	revenue numeric
1	Female	75191
2	Male	157890

SQL result — revenue by gender

Revenue by Gender

Male \$157,890 (68%) vs Female \$75,191 (32%)

Per-Customer Spend

Male $\approx$ \$59.5 • Female $\approx$ \$60.2

Insight

The male lead is driven by a larger customer count — women actually spend marginally more per head. Acquiring more female customers is an untapped lever.

Subscription & Loyalty — Biggest Opportunity

```
50 -- Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count
51 -- of each segment.
52 with customer_type as (
53   select customer_id, previous_purchases,
54   case
55     when previous_purchases = 1 Then 'New'
56     when previous_purchases Between 2 and 10 then 'Returning'
57     else 'loyal'
58   End as customer_segment
59   from customers
60 )
61
62 select customer_segment, count(*) as "Number of Customers"
63 from customer_type
64 group by customer_segment
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Coat	49.00
3	Sneakers	49.00
4	Sweater	48.00
5	Pants	47.00

SQL result — customer segmentation

Key Figures

- Subscriber vs non-subscriber spend: \$59.49 vs \$59.87
- Segments: Loyal 3,116 · Returning 701 · New 83
- Repeat buyers (5+): 958 subscribe vs 2,518 don't (~28%)

Insight

80% of customers are loyal repeat buyers, yet only ~27% subscribe — and subscription doesn't lift spend. Converting loyal non-subscribers is the win.

Products, Discounts & Shipping

Top-Rated Products

- Gloves 3.86
- Sandals 3.84
- Boots 3.82
- Hat 3.80
- Skirt 3.78

Best-Sellers by Category

- Jewelry (Accessories)
- Pants & Blouse (Clothing)
- Sandals (Footwear)
- Jacket (Outerwear)

Most Discount-Reliant

- Hat 50%
- Coat 49%
- Sneakers 49%
- Sweater 48%
- Pants 47%

Shipping

- Express \$60.48 vs Standard \$58.46
- 839 customers spent above average while using a discount

Insight Some products only sell with heavy discounts (margin review advised); faster shipping correlates with modestly higher spend.

Recommendations

1

Grow subscriptions

73% aren't subscribed and it doesn't lift spend — redesign the offer with real value.

2

Convert loyal non-subscribers

Target repeat buyers (80% of base) who don't subscribe.

3

Acquire more female customers

Women are 32% of base but spend slightly more per order.

4

Audit discount-reliant items

Review margins on Hat, Coat, Sneakers (~49–50% discounted).

5

Reinforce Clothing

The core category (~45%); investigate Outerwear's low share.

6

Feature top-rated products

Promote best-rated & best-sellers; consider premium pricing.

7

Invest in faster delivery

Express buyers spend modestly more.

Tools & Links

Tech Stack

- Python — pandas, Jupyter Notebook
- PostgreSQL — pgAdmin, SQLAlchemy + psycopg2
- Tableau — live database connection

Links

GitHub repository: github.com/Manas463/customer-shopping-behavior-analysis

Live dashboard: [View the interactive dashboard on Tableau Public →](#)

